
Flow Sampling: Learning to Sample from Unnormalized Densities via Denoising Conditional Processes

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Abstract

Sampling from unnormalized densities is analogous to the generative modeling problem, but the target distribution is defined by a known energy function instead of data samples. Because evaluating the energy function is often costly, a primary challenge is to learn an efficient sampler. We introduce *Flow Sampling*, a framework built on diffusion models and flow matching for the data-free setting. Our training objective is conditioned on a noise sample and regresses onto a *denoising* diffusion drift constructed from the energy function. In contrast, diffusion models’ objective is conditioned on a data sample and regresses onto a *noising* diffusion drift. We utilize the interpolant process to minimize the number of energy function evaluations during training, resulting in an efficient and scalable method for sampling unnormalized densities. Furthermore, our formulation naturally extends to Riemannian manifolds, enabling diffusion-based sampling in geometries beyond Euclidean space. We derive a closed-form formula for the conditional drift on constant curvature manifolds, including hyperspheres and hyperbolic spaces. We evaluate Flow Sampling on synthetic energy benchmarks, small peptides, large-scale amortized molecular conformer generation, and distributions supported on the sphere, demonstrating strong empirical performance.

1. Introduction

Many problems in the computational sciences require sampling from high-dimensional probability distributions that are specified only up to a normalization constant (Frenkel &

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Smit, 2023; Noé et al., 2019; Gabrié et al., 2022),

$$q(x) = \frac{\exp(r(x))}{Z}, \quad Z = \int_{\mathbb{R}^d} \exp(r(x)) dx < \infty. \quad (1)$$

The target distribution is q , and it is defined by the unnormalized log density $r(x)$, which is referred to as the *reward function* in reinforcement learning, or the *negative energy* in physical the sciences, computational chemistry and biology. One can typically evaluate $r(x)$ and its gradient $\nabla r(x)$, often at high computational cost, but does not have access to samples from q .

Markov chain Monte Carlo (MCMC) methods (Hastings, 1970; Neal, 2001), including Langevin dynamics (Roberts & Tweedie, 1996; Roberts & Rosenthal, 1998), converge asymptotically to the target distribution but often mix slowly and generate samples sequentially. This limits their applicability in settings such as materials design (Barroso-Luque et al., 2024) that require fast, reusable samplers across many target instances. This motivates the development of *amortized* sampling methods that replace long sequential simulations with learned sampling dynamics.

In this work, we introduce *Flow Sampling*, a diffusion-based framework for learning amortized samplers directly from unnormalized density functions. Flow Sampling learns to sample the target distribution using a conditional denoising process constructed from the energy function, such that its marginal distribution matches the target by design. It utilizes the known interpolant process and a detached state of the model to obtain reusable gradients of the negative energy which reduces considerably the required number of energy function evaluation. In contrast, existing diffusion-based methods (Phillips et al., 2024; De Bortoli et al., 2024; Akhound-Sadegh et al., 2024b) learn to sample the target distribution through Monte Carlo corrections, such as importance weighting, resampling, or auxiliary MCMC. While principled, these approaches rely on stochastic estimation and repeated energy function evaluations, which can significantly increase computational cost.

More recently, diffusion samplers based on stochastic optimal control or Schrödinger bridge problems have emerged. They learn the diffusion dynamics by optimizing divergences over diffusion path measures (Zhang & Chen, 2022b;

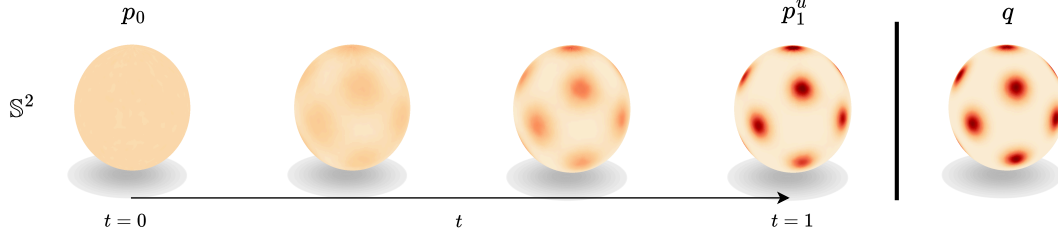


Figure 1. Flow Sampling is the first diffusion sampler able to perform sampling from an unnormalized density on Riemannian manifolds. The figure depicts a learned diffusion process using iterative Flow Sampling from a mixture of spherical von Mises–Fisher distributions.

Havens et al., 2025; Liu et al., 2025a). Although theoretically well grounded, this perspective requires characterizing optimal controls or bridges, leading to complex training schemes and auxiliary networks. By building on the diffusion models (Ho et al., 2020; Song et al., 2021) and flow matching (Lipman et al., 2023; Liu et al., 2023; Albergo et al., 2023) formulation, we alleviate these restrictions, resulting in a flexible design space and an easy to implement method. Despite the non-optimal control or coupling, in practice Flow Sampling cuts training cost by at least half on the large-scale amortized molecular conformer generation benchmarks (Havens et al., 2025). Moreover, the Flow Sampling naturally extends to Riemannian manifolds. Similar to Riemannian flow matching (Chen & Lipman, 2023), we replace the affine interpolant with a geodesic interpolant, and furthermore we derive a closed-form formula for the conditional drift on constant curvature manifolds, which includes the hypersphere and hyperbolic spaces.

Our main contributions:

1. We introduce Flow Sampling, a principled framework for learning diffusion samplers from unnormalized densities by matching conditional denoising diffusion drifts.
2. We derive a simple closed-form regression target that reuses endpoint energy gradients along the interpolant process, leading to an efficient replay-buffer training algorithm summarized in Algorithm 1.
3. We extend diffusion-based samplers to geometries beyond Euclidean spaces. In particular, we derive a closed-form formula for the conditional denoising diffusion drift on constant curvature Riemannian manifolds, including the hypersphere and the hyperbolic spaces.
4. We demonstrate state-of-the-art performance on synthetic energy benchmarks and amortized molecular conformer generation, while reducing simulation training cost by 4–8 times.

2. Preliminaries: Flow Matching

A flow process $(X_t)_{0 \leq t \leq 1}$ is defined by a *velocity field* $v : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ and a boundary condition,

$$dX_t = v_t(X_t) dt, \quad X_0 \sim p_0, \quad (2)$$

where the *source distribution* p_0 is the marginal of the process at time $t = 0$. The *probability path* p_t is the marginal of the process at time $0 \leq t \leq 1$, and it is given by the *continuity equation*,

$$\frac{\partial}{\partial t} p_t(x) + \nabla [p_t(x) v_t(x)] = 0. \quad (3)$$

Given a dataset of i.i.d. samples from a target distribution q on $\mathbb{R}^d$, and an easy to sample source distribution p_0 , the goal of flow matching (FM) (Lipman et al., 2023; Liu et al., 2023; Albergo & Vanden-Eijnden, 2023) is to learn a velocity field v_t^θ such that the marginal of X_t interpolate between the source and the target, i.e.,

$$X_0 \sim p_0, \quad X_1 \sim p_1 \equiv q. \quad (4)$$

To learn v_t^θ , FM uses a supervising flow process conditioned at time $t = 1$. Given $X_1 = x_1$, the conditional velocity is,

$$v_{t|1}(x|x_1) = \frac{\dot{\sigma}_t}{\sigma_t} (x - \alpha_t x_1) + \dot{\alpha}_t x_1. \quad (5)$$

The ODE (2) for the conditional flow (5) is solved by,

$$X_t = \sigma_t X_0 + \alpha_t x_1, \quad X_0 \sim p_0. \quad (6)$$

The marginal of the *interpolant* (6) X_t is the *conditional probability path* that is given by the push forward formula

$$p_{t|1}(x|x_1) = \frac{1}{\sigma_t^d} p_0 \left(\frac{x - \alpha_t x_1}{\sigma_t} \right). \quad (7)$$

Finally, the training objective for the velocity is the FM loss

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E} \left[\|v_t^\theta(X_t) - v_{t|1}(X_t|X_1)\|^2 \right], \quad (8)$$

where $X_1 \sim q$ sample is obtained from the dataset, and $X_t \sim p_{t|1}$ is sampled using the interpolant (6).

Indeed, the generated probability path by the optimal v_t^θ is

$$p_t(x) = \mathbb{E} [p_{t|1}(x|X_1)], \quad (9)$$

that equals the source, p_0 and target q at $t = 0$ and $t = 1$.

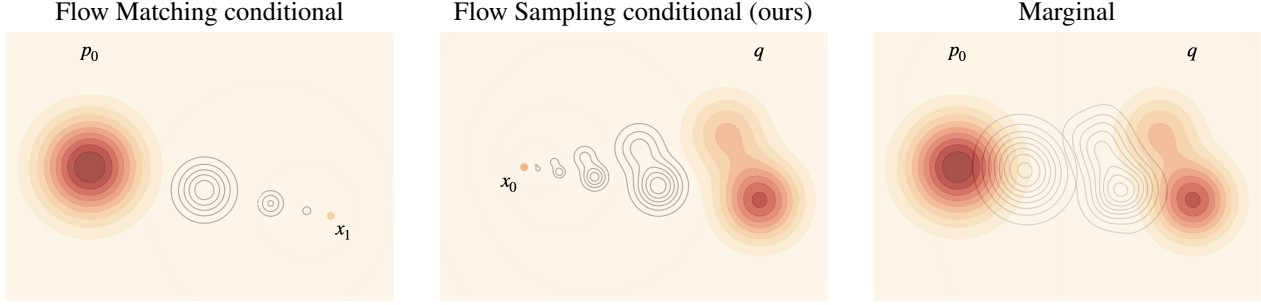


Figure 2. (left) Flow matching conditional probability path is marginal of a noising process conditioned on a data point x_1 . (middle) Flow sampling conditional probability path is a marginal of a denoising process conditioned on a noise point x_0 . (right) The probability of the marginal process is the same in both cases.

3. Flow Sampling

In contrast to flow matching, we are interested in the data free settings, but our goal is similar. Given a score function of the target $\nabla \log q(x_1) = \nabla r(x_1)$, and an easy to sample source distribution p_0 , our goal is to learn a diffusion process $(X_t)_{0 \leq t \leq 1}$ such that its probability path interpolates between source p_0 and target $p_1 \equiv q$.

Diffusion process A diffusion process is defined by a drift $u : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$, a diffusion coefficient $g_t : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$, and a boundary condition,

$$dX_t = u_t(X_t) dt + g_t dB_t, \quad X_0 \sim p_0. \quad (10)$$

where B_t is the Brownian motion. The probability path p_t of a diffusion process is given by the Fokker–Planck equation,

$$\frac{\partial}{\partial t} p_t(x) + \nabla [p_t(x) u_t(x)] = \frac{g_t^2}{2} \nabla^2 p_t(x). \quad (11)$$

We recall the standard correspondence between deterministic transport and a diffusion with the same probability path.

Proposition 3.1. *Let v_t be a velocity that defines the flow process (2) with a marginal p_t . Then the drift*

$$u_t(x) = v_t(x) + \frac{g_t^2}{2} \nabla \log p_t(x), \quad (12)$$

defines a diffusion process (10) with the same marginal p_t .

Proposition 3.1 is a known result (Anderson, 1982); we defer the proof to the appendix. This observation makes explicit how score information enters the drift, which is exactly the mechanism we will exploit to inject target information.

Key Idea Given a data sample $X_1 \sim q$, the conditional flow (5) which is the supervision for the FM loss (8), is independent of the target distribution q . Thus, all learning signals for the target q come through marginalization.

In data-free settings, we don’t have access to data samples $X_1 \sim q$, hence we cannot marginalize over the target q . Our

key observation is that we can condition on samples from the source $X_0 \sim p_0$, move all the signal about the target q to a supervising denoising diffusion process conditioned at time $t = 0$ by X_0 , and marginalize over the source p_0 .

We use a conditional probability path defined as a push forward of the target distribution

$$p_{t|0}(x|x_0) = \frac{1}{\alpha_t^d} q\left(\frac{x - \sigma_t x_0}{\alpha_t}\right). \quad (13)$$

This is in contrast to flow matching and diffusion models (Ho et al., 2020) that pushes the source distribution (7). Still, in both cases the marginal probability path is the same,

$$\mathbb{E}[p_{t|0}(x|X_0)] = p_t(x) = \mathbb{E}[p_{t|1}(x|X_1)]. \quad (14)$$

The difference between pushing forward the source p_0 or the target q is illustrated in Figure 2. Notably, the former is independent of the target, and the latter is a scale and shift of the target.

Supervising drift Importantly, we identify two processes that generate the conditional path in (13). The first, is defined by a conditional velocity similar to FM (5) but conditioned on time $t = 0$. That is,

$$v_{t|0}(x|x_0) = \frac{\dot{\alpha}_t}{\alpha_t} (x - \sigma_t x_0) + \dot{\sigma}_t x_0, \quad (15)$$

which generates the probability path $p_{t|0}$ (13) since they satisfy the continuity equation (3). The second, is defined by the conditional drift

$$u_{t|0}(x|x_0) := v_{t|0}(x|x_0) + \frac{g_t^2}{2} \nabla \log p_{t|0}(x|x_0), \quad (16)$$

which by Proposition (3.1), generates the same conditional probability path $p_{t|0}$ (13).

The two processes are important. The second process, i.e., the drift (16) allows to sample $X_1 \sim q$ given $X_0 = x_0$ using simulation, and regressing onto the drift $u_{t|0}$ suffices to learn a sampler of q . The first process, i.e., the velocity (15) is solved in closed form by the interpolant (6) and allows efficient reuse of existing samples $X_1 \sim q$ with simulation free sampling of $p_{t|0}$, significantly reducing training cost.

Training objective Similar to FM, we use the marginalization trick (Lipman et al., 2024) and train our drift u^θ by regressing onto the conditional drift $u_{t|0}$ (16). This yields a *diffusion matching* (DM) objective

$$\mathcal{L}_{\text{DM}}(\theta) = \mathbb{E} \left[\|u^\theta(X_t) - u_{t|0}(X_t|X_0)\|^2 \right], \quad (17)$$

where $X_0 \sim p_0$ and $X_t \sim p_{t|0}$. The conditional $p_{t|0}$ is sampled using the flow process (2) defined by $v_{t|0}$ (15) which can be done efficiently since it is solved by the interpolant,

$$X_t = \sigma_t X_0 + \alpha_t X_1, \quad X_0 \sim p_0, X_1 \sim q. \quad (18)$$

Proposition 3.2. *Let $X_0 = x_0$, and $v_{t|0}$ the conditional velocity (15) that generates $p_{t|0}$ the conditional probability path (13). Then, for every $g_t \geq 0$ and $X_1 \sim q$,*

$$u_{t|0}(X_t|x_0) = \dot{\alpha}_t X_1 + \dot{\sigma}_t x_0 + \frac{g_t^2}{2\alpha_t} \nabla r(X_1). \quad (19)$$

where X_t is the interpolant (18) defined by $v_{t|0}$.

Proof of Proposition 3.2 is in Appendix E. This proposition is extremely useful, since it implies that for every sample from the target $X_1 \sim q$, the score $\nabla \log q(X_1) = \nabla r(X_1)$ needs to be evaluated only once, to evaluate $u_{t|0}$, the conditional drift (19), along X_t the interpolant (18) for all $t \in [0, 1]$ and $X_0 \sim p_0$. Hence, we can use a *replay buffer* $\mathcal{B}$ to store the pairs $X_1, \nabla r(X_1)$ and reuse them during training. This improves training efficiency significantly in cases where the reward is costly to compute.

Linear process We train our model using the known linear scheduler (Lipman et al., 2023; Liu, 2022),

$$\alpha_t = t, \quad \sigma_t = 1 - t, \quad g_t^2 = 2\gamma t, \quad \gamma > 0. \quad (20)$$

The squared diffusion coefficient is chosen $g_t^2 \propto \alpha_t$ to account for the possible singularity at time $t = 0$ as implied by Proposition (3.2). This yields a stable target for the regression, and γ is a hyper-parameter that is used to further regularize the target during training.

3.1. Diffusion matching with the Flow Sampling loss

Sampling the target $X_1 \sim q$ using simulation with the conditional drift (16) requires repeated evaluations of the energy gradient $\nabla r(x)$, which we want to avoid.

Fixed point iteration To overcome this, we propose a fixed point iteration training algorithm. We defined X_1^θ to be samples from the detached current state of the model, i.e., $\bar{\theta} = \text{stopgrad}(\theta)$, using the Euler–Maruyama algorithm,

$$X_{t+h}^\theta = X_t^\theta + h u_t^\theta(X_t^\theta) + \sqrt{2\gamma t h} Z_t, \quad (21)$$

where $Z_t \sim \mathcal{N}(0, I)$. Then, by minimizing the DM loss (17), but sampling $X_1^\theta \sim p_1^\theta$ instead of $X_1 \sim q$ during training we obtain our flow sampling objective.

Algorithm 1 Flow sampling training

```
# model_fn - trainable drift model
# solver_fn - Euler--Maruyama solver
# score_fn - gradient of the reward
# buffer - replay buffer
# gam - diffusion coefficient
# bz, shape - batch size and data shape
for _ in range(outer_loops):
    # exploration phase
    x_0 = randn(bz, *shape)
    with no_grad():
        x_1 = solver_fn(model_fn, gam, x_0)
        score_1 = score_fn(x_1)
    buffer.push(x_1, score_1)

    # optimization phase
    for _ in range(inner_loops):
        x_1, score_1 = buffer.sample(bz)
        x_0 = randn(bz, *shape)
        t = rand(bz, *[1]*len(shape))

        x_t = (1-t)*x_0 + t*x_1
        u_t = x_1 - x_0 + gam*score_1
        pred = model_fn(x_t, t)
        loss = mse_loss(pred, u_t)

    loss.backward()
    optimizer.step()
    optimizer.zero_grad()
```

Flow Sampling Objective We train u_t^θ by regressing onto a closed-form target (19), and linear process (20). The target distribution is approximated with samples of the current detached state of the model $X_1^\theta \sim p_1^\theta$.

$$\mathcal{L}_{\text{FS}}(\theta) = \mathbb{E} \left[\|u_t^\theta(X_t) - (X_1^\theta - X_0 + \gamma \nabla r(X_1^\theta))\|^2 \right], \quad (22)$$

where X_t is the linear interpolant process

$$X_t = (1-t)X_0 + tX_1^\theta, \quad X_0 \sim p_0, X_1^\theta \sim p_1^\theta. \quad (23)$$

We use the same alternating scheme as Adjoint Sampling (Havens et al., 2025), which consists of two phases:

1. **Exploration phase:** Using the current detached model u^θ , we draw a batch of samples $X_1^\theta \sim p_1^\theta$ with the Euler–Maruyama solver, evaluate the score $\nabla r(X_1^\theta)$, and store the pairs in the replay buffer $\mathcal{B} \leftarrow \mathcal{B} \cup \{X_1^\theta, \nabla r(X_1^\theta)\}$.
2. **Optimization phase:** Fixing the state of the replay buffer $\mathcal{B}$, we iterate over it $\{X_1^\theta, \nabla r(X_1^\theta)\} \sim \mathcal{B}$, and perform multiple gradient steps on $\mathcal{L}_{\text{FS}}(\theta)$ (22).

The two phases are repeated until convergence. The pseudo code for our training method is given in Algorithm 1, where a Gaussian source $p_0 = \mathcal{N}(0, I)$ is assumed. The number of energy gradient $\nabla r(x)$ evaluations equals the number of $X_1^\theta \sim p_1^\theta$ acquired during training (Havens et al., 2025).

4. Extension to Riemannian Manifolds

A strong advantage of our flow sampling formulation for training diffusion samplers (Akhound-Sadeh et al., 2024a; Vargas et al., 2023a; Havens et al., 2025) is that it naturally extends diffusion samplers to non-Euclidean geometries.

This section describes how to adapt flow sampling to Riemannian Manifolds. Observing our FS loss (22) we identify three key requirements: (i) simulation of diffusion process on manifolds (Huang et al., 2022) for sampling, (ii) an interpolant process (18) on manifolds (Chen & Lipman, 2023), and (iii) a conditional drift (19) on manifolds that generate the same marginal as the interpolant.

Setup Let an ambient space $\mathcal{A} = \mathbb{R}^{d+1}$ equipped with a metric $\langle \cdot, \cdot \rangle_\Sigma$ given by $\Sigma \in \mathbb{R}^{(d+1) \times (d+1)}$ an invertible symmetric matrix. We assume a complete, connected, smooth Riemannian manifold with a constant curvature κ given by

$$\mathcal{M} = \left\{ x \in \mathcal{A} \mid \langle x, x \rangle_\Sigma = \frac{1}{\kappa} \right\} \quad (24)$$

Note, in case $\mathcal{M}$ has more than one connected component, we treat each one separately. The tangent space $T_x \mathcal{M}$ at a point $x \in \mathcal{M}$ is defined as

$$T_x \mathcal{M} = \{ w \in \mathcal{A} \mid \langle w, x \rangle_\Sigma = 0 \}. \quad (25)$$

Diffusion on manifolds As in the Euclidean case, a diffusion process is defined by a drift $u_t(x) \in T_x \mathcal{M}$, and a boundary condition,

$$dX_t = u_t(X_t) dt + \sqrt{2\gamma t} P_{X_t}^\perp \circ dB_t, \quad X_0 \sim p_0. \quad (26)$$

where, B_t is the standard Brownian motion in the ambient space $\mathcal{A}$, the diffusion coefficient is $g_t := \sqrt{2\gamma t}$, and $P_x^\perp$ is an orthogonal projection defined for any $x, w \in \mathcal{A}$ as

$$P_x w = \frac{\langle x, w \rangle_\Sigma}{\langle x, x \rangle_\Sigma} x, \quad P_x^\perp w = w - P_x w. \quad (27)$$

Thus, $P_{X_t}^\perp \circ dB_t \in T_{X_t} \mathcal{M}$. The marginal of the process p_t is defined by the Riemannian Fokker-Planck equation

$$\partial_t p_t = -\text{div}_{\mathcal{M}}(p_t u_t) + \gamma t \Delta_{\mathcal{M}} p_t, \quad (28)$$

where $\text{div}_{\mathcal{M}}$ and $\Delta_{\mathcal{M}}$ are the divergence and Laplacian operators (resp.) generalized to manifolds.

Lastly, the Euler–Maruyama solver on the manifold is

$$X_{t+h} = \exp_{X_t} \left(h u_t(X_t) + \sqrt{2\gamma t h} P_{X_t}^\perp Z_t \right), \quad (29)$$

where $Z_t \sim \mathcal{N}(0, I)$, and $\exp_x : T_x \mathcal{M} \rightarrow \mathcal{M}$ is the exponential map of $\mathcal{M}$. For our set of constant curvature manifolds (24), the exponential map can be solved in closed form (Chen & Lipman, 2023) that is dependent on Σ and κ .

Geodesic Interpolant Given $X_0 = x_0 \in \mathcal{M}$ and $X_1 \sim q$, similar to Riemannian flow matching (RFM) (Chen & Lipman, 2023), instead of an interpolant (18) between X_0 and X_1 , we use the geodesic,

$$X_t = \phi_t(X_1|x_0) = \exp_{X_1} \left[(1-t) \log_{X_1}(x_0) \right], \quad (30)$$

where $\log_x : \mathcal{M} \rightarrow T_x \mathcal{M}$ is the logarithmic map which can be solved in closed form for the set of manifolds we assumed. The marginal of the geodesic interpolant is the conditional probability path,

$$p_{t|0}(x|x_0) = q(\phi_{t|0}^{-1}(x|x_0)) \left| \det_{\mathcal{M}} D\phi_{t|0}^{-1}(x|x_0) \right|. \quad (31)$$

Conditional drift Similar to the Euclidean case, we find a conditional drift $u_{t|0}$ that generate the conditional path (31) $p_{t|0}$. Evaluated at the geodesic interpolant (30), the drift is

$$u_{t|0}(X_t|x_0) = \dot{X}_t + \gamma t \nabla_{\mathcal{M}} \log p_t(X_t|x_0), \quad (32)$$

where $\dot{X}_t = \frac{d}{dt} \phi_t(X_1|x_0)$ is the derivative w.r.t. time of the geodesic, and the conditional score decomposes to

$$\nabla_{\mathcal{M}} \log p_{t|0}(X_t|x_0) = (J_t^{-1})^* \nabla_{\mathcal{M}} r(X_1) \quad (33)$$

$$- \nabla_{\mathcal{M}} \log \left| \det_{\mathcal{M}}(J_t) \right|, \quad (34)$$

where $J_t = D\phi_t(X_1|x_0)$ is the Jacobian of the geodesic, and $(J_t^{-1})^*$ is its inverse adjoint.

Proposition 4.1. *For Riemannian manifold $\mathcal{M}$ defined in (24) with a metric Σ and constant curvature κ , the Jacobian of the geodesic w.r.t. X_1 is*

$$J_t = t T_{X_1 \rightarrow X_t} P_{\dot{X}_1} + c_t(X_1, x_0) T_{X_1 \rightarrow X_t} P_{\dot{X}_1}^\perp,$$

where $T_{X_1 \rightarrow X_t} : T_{X_1} \mathcal{M} \rightarrow T_{X_t} \mathcal{M}$ is the parallel transport, $c_t(X_1, x_0)$ is a time dependent scaling,

$$c_t(X_1, x_0) = \begin{cases} \frac{\sin(t\omega_1\sqrt{\kappa})}{\sin(\omega_1\sqrt{\kappa})} & \text{if } \kappa > 0 \\ \frac{\sinh(t\omega_1\sqrt{|\kappa|})}{\sinh(\omega_1\sqrt{|\kappa|})} & \text{if } \kappa < 0 \end{cases}, \quad (35)$$

and $\omega_1 = \left\| \dot{X}_1 \right\|_\Sigma$ the geodesic distance of X_1 and x_0 .

Proposition (4.1) allows us to compute the conditional drift (32) in closed form. As it is composed of rank-1 matrices, it is efficient to evaluate. Its proof is based on results in Jacobi fields (Lee, 2018), and given in Appendix E.

Riemannian Flow Sampling objective The model u_t^θ is trained by regressing onto $u_{t|0}$ the conditional drift (32) that is given in closed form in Proposition 4.1.

$$\mathcal{L}_{\text{RFS}}(\theta) = \mathbb{E} \left[\|P_{X_t}^\perp u_t^\theta(X_t) - u_{t|0}(X_t|X_0)\|_\Sigma^2 \right] \quad (36)$$

where X_t is the geodesic interpolant (30), and the target distribution is approximated with $X_1^\theta \sim p_1^\theta$ samples of the current detached state of the model using the Riemannian Euler-Maruyama solver (29). Optimization is done using the adjusted to manifold version of Algorithm 1.

4.1. Flow sampling on a hyper-sphere

As an example, we consider the case of a hyper-sphere $\mathcal{M} = \mathbb{S}^d$ given by setting the metric and the curvature to

$$\Sigma = I, \quad \kappa = 1. \quad (37)$$

Given $X_0 = x_0 \in \mathbb{S}^d$, and $X_1 \sim q$, the geodesic distance is

$$\omega_1 = \|\log_{X_1}(x_0)\| = \arccos(X_1^\top x_0). \quad (38)$$

The derivative w.r.t. time of the geodesic interpolant (30) at time $t = 1$ is

$$\dot{X}_1 = -\log_{X_1}(x_0) = \omega_1 \frac{\cos \omega_1 X_1 - x_0}{\sin \omega_1}, \quad (39)$$

and, the geodesic interpolant (30) is the SLERP function,

$$X_t = \cos((1-t)\omega_1) X_1 - \sin((1-t)\omega_1) \frac{\dot{X}_1}{\omega_1}, \quad (40)$$

and its derivative w.r.t. to time t is

$$\dot{X}_t = \omega_1 \sin((1-t)\omega_1) X_1 + \cos((1-t)\omega_1) \dot{X}_1. \quad (41)$$

The parallel transport from X_1 to X_t on the hyper-sphere is the Householder reflection about the midpoint $X_1 + X_t$,

$$T_{X_1 \rightarrow X_t} = I - 2P_{X_1 + X_t}. \quad (42)$$

Since the parallel transport $T_{X_1 \rightarrow X_t}$ is an orthogonal operator, using Proposition 4.1 the inverse Jacobian adjoint is

$$(J_t^{-1})^* = \frac{1}{t} T_{X_1 \rightarrow X_t} P_{\dot{x}_1} + \frac{\sin(\omega_1)}{\sin(t\omega_1)} T_{X_1 \rightarrow X_t} P_{\dot{X}_1}^\perp. \quad (43)$$

This implies the correction to the conditional score is

$$\nabla_{\mathcal{M}} \log \det(J_t) = (d-1)(t \cot t\omega_1 - \cot \omega_1) \frac{\dot{X}_1}{\omega_1}. \quad (44)$$

Hence we have all terms in the conditional drift (32) in closed form. Importantly, both the projection operator (27) and the parallel transport operator (42) include only rank 1 operators, thus are computationally efficient.

5. Related Works

Learning-Augmented Classical Samplers MCMC and SMC methods provide asymptotically exact samples from unnormalized distributions, but can mix slowly or require many target evaluations in high-dimensional, multi-modal settings. Learning-augmented variants replace hand-designed proposals with normalizing flows or learnable stochastic dynamics (Albergo et al., 2019; Arbel et al., 2021; Gabri   et al., 2022; Matthews et al., 2022; Albergo & Vanden-Eijnden, 2025; Holderrieth et al., 2025), typically requiring importance weighting or accept-reject corrections that drive up energy-evaluation cost.

Diffusion Samplers Diffusion samplers adapt score matching to unnormalized targets via auxiliary corrections (MCMC, importance weighting, or resampling), each inflating energy-evaluation cost (Phillips et al., 2024; De Bortoli et al., 2024; Chen et al., 2024). PITA (Akhound-Sadegh et al.) trains a temperature-annealed ladder of diffusion samplers, but requires high-temperature MCMC data to bootstrap. Off-policy variants (Malkin et al., 2023; Richter & Berner, 2024) use trajectory-level objectives over a replay buffer but typically require parameterizing the model in terms of the energy gradient (He et al., 2025). iDEM (Akhound-Sadegh et al., 2024b) instead estimates the target score by Monte Carlo along the noising path, incurring many energy evaluations per step to control variance.

Stochastic Optimal Control and Adjoint Sampling A closely related line of work formulates sampling from unnormalized distributions as a stochastic optimal control problem, closely connected to Schr  dinger bridges (Zhang & Chen, 2022a; Vargas et al., 2023b), leading to bridge-based diffusion samplers and transport objectives (Berner et al., 2024; Richter & Berner, 2024; Vargas et al., 2024; Chen et al., 2024; Havens et al., 2025; Liu et al., 2025a). (Nam et al., 2025) further incorporates molecular inductive bias through well-tempered collective-variable biasing. Flow Sampling is inspired by this line of work but departs from SOC formulations by avoiding optimal couplings, yielding a simpler training objective that supports non-memoryless noise schedules and arbitrary source distributions without requiring an additional corrector network. In Appendix D we show that with a Brownian-bridge supervising path and matching Gaussian source, the Flow Sampling and Adjoint Sampling regression targets agree in conditional expectation.

Reward-tilting and Fine-Tuning RL-style fine-tuning methods such as GRPO (Shao et al., 2024) and Flow-GRPO (Liu et al., 2025b) optimize a pretrained generative model toward high-reward samples via policy gradients, without explicitly targeting a particular distribution. A complementary line instead targets the reward-tilted distribution—the base reweighted by an exponential reward.

Table 1. Results for the synthetic energy function experiments. We report a geometric $\mathcal{W}_2$ metric Klein et al. (2024) and 1D energy histogram $E(\cdot)\mathcal{W}_2$ metric with respect to ground truth samples. See Section A.4 for more details. We use light and dark shading to denote best and 2nd best result respectively. [†] We include baseline measurements over random subsets of the ground truth samples. This can be considered the “optimal” value for that metric, which was not clear in previous instantiations of this benchmark. Flow Sampling XL results are omitted for DW4 due to minimal improvement. Mean and standard error are reported across 5 training runs.

Method	DW-4 ($N = 4, d = 2$)		LJ-13 ($N = 13, d = 3$)		LJ-55 ($N = 55, d = 3$)	
	$\mathcal{W}_2 \downarrow$	$E(\cdot)\mathcal{W}_2 \downarrow$	$\mathcal{W}_2 \downarrow$	$E(\cdot)\mathcal{W}_2 \downarrow$	$\mathcal{W}_2 \downarrow$	$E(\cdot)\mathcal{W}_2 \downarrow$
PIS (Zhang & Chen, 2022a)	0.68 $\pm$ 0.23	0.65 $\pm$ 0.25	1.93 $\pm$ 0.07	18.02 $\pm$ 1.12	4.79 $\pm$ 0.45	228.70 $\pm$ 131.27
DDS (Vargas et al., 2023b)	0.92 $\pm$ 0.11	0.90 $\pm$ 0.37	1.99 $\pm$ 0.13	24.61 $\pm$ 8.99	4.60 $\pm$ 0.09	173.09 $\pm$ 18.01
iDEM (Akhound-Sadegh et al., 2024b)	0.70 $\pm$ 0.06	0.55 $\pm$ 0.14	1.61 $\pm$ 0.01	30.78 $\pm$ 24.46	4.69 $\pm$ 1.52	93.53 $\pm$ 16.31
AS (Havens et al., 2025)	0.62 $\pm$ 0.06	0.55 $\pm$ 0.12	1.67 $\pm$ 0.01	2.40 $\pm$ 1.25	4.04 $\pm$ 0.05	30.83 $\pm$ 8.19
ASBS (Liu et al., 2025a)	0.43 $\pm$ 0.05	0.20 $\pm$ 0.11	1.59 $\pm$ 0.03	1.99 $\pm$ 1.01	4.00 $\pm$ 0.03	28.10 $\pm$ 8.15
Flow Sampling	0.36 $\pm$ 0.03	0.11 $\pm$ 0.04	1.58 $\pm$ 0.01	0.97 $\pm$ 0.53	3.98 $\pm$ 0.01	21.32 $\pm$ 0.63
Flow Sampling XL	—	—	1.57 $\pm$ 0.01	0.81 $\pm$ 0.26	3.96 $\pm$ 0.01	16.29 $\pm$ 0.93
Subsets of MCMC (ground truth) [†]	0.31 $\pm$ 0.01	0.10 $\pm$ 0.03	1.57 $\pm$ 0.01	0.35 $\pm$ 0.03	3.85 $\pm$ 0.05	0.57 $\pm$ 0.05

Concurrent to our work, Tilt Matching (Potapchik et al., 2025) regresses a velocity field that anneals from a base toward the tilted target using only reward evaluations, while Meta Flow Maps (Potapchik et al., 2026) amortize one-step posterior sampling across intermediate noise levels to enable steering and off-policy fine-tuning.

6. Experiments

6.1. Synthetic Energy Functions

We evaluate Flow Sampling on standard synthetic n -particle energy benchmarks (Köhler et al., 2020; Midgley et al., 2023; Klein et al., 2024; Akhound-Sadegh et al., 2024b): a 2D 4-particle Double-Well potential (DW-4), a 3D 13-particle Lennard–Jones system (LJ-13), and a 55-particle Lennard–Jones system (LJ-55). The states are particle positions $x \in \mathbb{R}^{n \times d}$, corresponding to $\text{shape} = (N, d)$ in Algorithm 1 and reward $r(x) = -E(x)$, where E is the energy defined by the potentials LJ or DW. Details in Appendix A. We report geometric and energy $\mathcal{W}_2$ distances against long-run MCMC reference samples (Klein et al., 2024).

Baselines We compare against iDEM (Akhound-Sadegh et al., 2024b), PIS (Zhang & Chen, 2022a), DDS (Vargas et al., 2023b), and the adjoint-based solvers AS (Havens et al., 2025) and ASBS (Liu et al., 2025a). All methods use a 5-layer EGNN (Satorras et al., 2021), with a 10-layer variant (Flow Sampling XL) evaluated for our method. We use the harmonic prior as in ASBS (Jing et al., 2023). Additionally, we make use of the zero-center-mass projected diffusion to handle translation invariance similarly to (Havens et al., 2025; Liu et al., 2025a).

Flow Sampling outperforms prior methods across all systems (Table 1), achieving the lowest $E(\cdot)\mathcal{W}_2$ and improved concentration in low-energy regions compared to the state-of-the-art ASBS (Liu et al., 2025a).

6.2. Sampling Conformers of Peptides

We evaluate Flow Sampling on two widely-studied peptide systems: 22-atom alanine dipeptide (Ala2) and 42-atom alanine tetrapeptide (Ala4), which are modeled with a classical force-field and implicit water solvation using the OpenMM library (Eastman et al., 2017). Notably, we sample Ala2 and Ala4 conformations in *full-atom* Cartesian coordinates, unlike prior sampling frameworks (Liu et al., 2025a; Zhang & Chen, 2022a) which relied on internal torsional coordinate representations and (Nam et al., 2026), which makes use of bespoke collective variables biases and pretraining data. As baseline, we take the ASBS configuration from (Nam et al., 2026), but skip their MD pretraining step. Both Flow Sampling and ASBS utilize the same $E(3)$ -equivariant PaiNN architecture (Schütt et al., 2021).

Results For Ala2, we follow prior works (Zhang & Chen, 2022a; Liu et al., 2025a) and report both the 1D KL divergence of each torsional dihedral angles (ϕ, ψ) and the Jensen-Shannon divergence (JSD) of the 2D joint against ground truth samples generated via long molecular dynamics (MD) simulation. We also report the energy $\mathcal{W}_2$ to measure concentration of samples about low energy. Flow sampling performs favorably compared to ASBS on all metrics shown in Table 2 and accurately recovers the ground truth density of the backbone dihedral angles shown by Ramachandran plots in Figure 4. Additionally, Table 2 shows that Flow Sampling appears to degrade in performance more gracefully than ASBS as the training NFE decreases, where ASBS fails at NFE 128 and 64, as indicated by the large energy- $\mathcal{W}_2$ metric. Lower model NFE can accelerate training and inference, which we discuss in greater detail later in Section 6.5. Additional experiment details energy histogram plots can be found in Section B.

For Ala4, we report a qualitative study of mode coverage and sample concentration via a 3D scatter plot over torsion

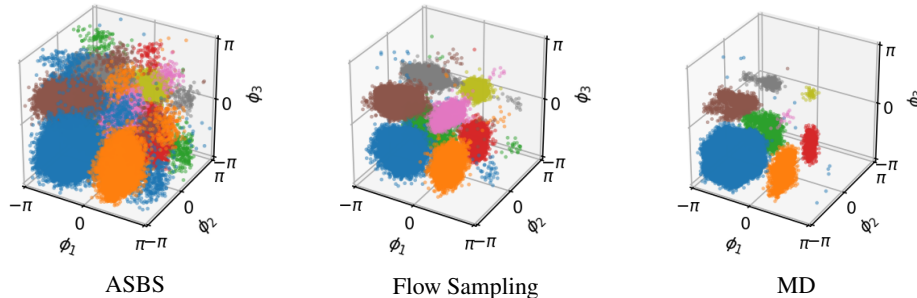


Figure 3. Ala4 molecular structures represented by torsion coordinates (ϕ_1, ϕ_2, ϕ_3) with 2^3 distinct modes.

Table 2. Results for ALA2. We report D_{KL} on 1-D marginals for ϕ and ψ , Jensen–Shannon divergence (JSD) on the joint distribution of torsion angles (ϕ, ψ) , and an energy-histogram $\mathcal{W}_2$ metric with respect to ground-truth samples. Light and dark shading denote the best and second-best method respectively.

Method	NFE (Train)	D_{KL} 1-D ($\downarrow$)		(ϕ, ψ) joint JSD ($\downarrow$)	Energy $\mathcal{W}_2$ ($\downarrow$)
		ϕ	ψ		
ASBS	1024	0.504 $\pm$ 0.006	0.726 $\pm$ 0.300	0.242 $\pm$ 0.042	8.650 $\pm$ 0.371
	256	0.412 $\pm$ 0.038	0.631 $\pm$ 0.419	0.233 $\pm$ 0.090	7.290 $\pm$ 2.625
	128	0.663 $\pm$ 0.460	0.550 $\pm$ 0.267	0.285 $\pm$ 0.158	1.0e7 $\pm$ 1.0e7
	64	1.127 $\pm$ 0.002	0.821 $\pm$ 0.006	0.441 $\pm$ 0.002	6.0e7 $\pm$ 8.8e5
Flow Sampling	1024	0.031 $\pm$ 0.004	0.008 $\pm$ 0.002	0.018 $\pm$ 0.001	0.637 $\pm$ 0.267
	256	0.235 $\pm$ 0.067	0.211 $\pm$ 0.035	0.118 $\pm$ 0.023	3.266 $\pm$ 0.254
	128	0.267 $\pm$ 0.023	0.344 $\pm$ 0.057	0.152 $\pm$ 0.021	3.579 $\pm$ 0.311
	64	0.316 $\pm$ 0.058	0.523 $\pm$ 0.006	0.190 $\pm$ 0.005	4.113 $\pm$ 0.023

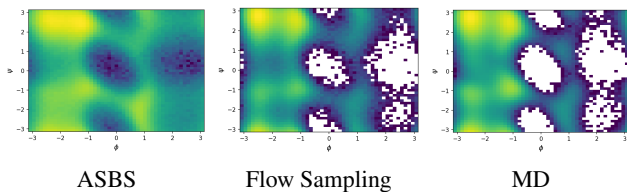


Figure 4. Ala2 Ramachandran plots over 10^6 samples.

angles (ϕ_1, ϕ_2, ϕ_3) , which gives rise to $2^3 = 8$ metastable modes corresponding to the possible combinations of the three backbone torsional states. As depicted in Figure 3, Flow Sampling covers all 8 modes observed in the MD reference samples. The samples are also more tightly concentrated around the MD-supported torsional regions than those generated by ASBS, suggesting improved alignment with the relevant conformational landscape. This indicates that Flow Sampling can capture the multimodal structure of Ala4 while maintaining better sample concentration in physically plausible regions.

6.3. Amortized Conformer Generation

We evaluate Flow Sampling on the large-scale amortized molecular conformer generation benchmark of Havens et al. (2025), where the target is a conditional Boltzmann distribution $r(x|g) = -\frac{1}{\tau} E(x|g)$, with energy given by the learned eSEN force field (Fu et al., 2025) and conditioning g that

provides the molecule topology (atom types and bonds).

For a molecule with N atoms, a conformer is represented as atomic coordinates $x \in \mathbb{R}^{N \times 3}$, corresponding to $\text{shape} = (N, 3)$ in Algorithm 1. Generation is conditioned on a fixed molecular graph g , as input to the drift network and energy. The model therefore defines a conditional diffusion process over atomic positions.

We train on 24,775 molecular topologies from SPICE (Eastman et al., 2023) and evaluate on 80 held-out SPICE molecules and a generalization dataset of 80 GEOM-DRUGS molecules (Axelrod & Gomez-Bombarelli, 2022), following the protocol of Havens et al. (2025). RDKit (Landrums, 2006) is a widely used fast conformer generation tool that provides a strong baseline. For more details see (Havens et al., 2025).

6.4. Von Mises-Fisher Mixtures on $\mathbb{S}^2$

Directional and rotational data arise in applications such as robotics and graphics, and are naturally represented on spherical domains (Bullo & Lewis, 2005; Bronstein et al., 2021). To validate the Riemannian Flow Sampling framework developed in Section 4.1, we consider sampling on the two-dimensional unit sphere

$$\mathbb{S}^2 = \{x \in \mathbb{R}^3 : \|x\|_2 = 1\}.$$

We construct a multimodal target distribution given by a mixture of von Mises–Fisher (vMF) (Banerjee et al., 2005) components, the spherical analogue of Gaussians, with density

$$f_{\text{vMF}}(x) \propto \exp(\kappa \mu^\top x)$$

parameterized by mean directions $\mu \in \mathbb{S}^2$ and concentration $\kappa \geq 0$. We then construct a mixture distribution

$$q(x) \propto \sum_{k=1}^K w_k f_{\text{vMF}}(x; \mu_k, \kappa_k), \quad (45)$$

where the mixture components are evenly weighted ($w_k = 1/K$) and the mean directions $\{\mu_k\}_{k=1}^K$ are chosen to lie along the coordinate axes and diagonal directions of $\mathbb{R}^3$.

Table 3. Results on large-scale amortized conformer generation, evaluated on two test sets, SPICE and GEOM-DRUGS using the same 12-layer EGNN architecture across all methods. We report the coverage recall and precision (%) and Absolute Mean RMSD (AMR) for the threshold $1.25\text{\AA}$ and standard deviation across test molecules. See Section 6.3 and Appendix C for details. Precision coverage was not reported in ASBS (Liu et al., 2025a). [†] Re-implementation with logit normal time weighting. * Auxiliary network roughly doubles the number of network evaluation during optimization.

Method	NFE (Train)	SPICE				GEOM-DRUGS			
		Recall		Precision		Recall		Precision	
		Cov. $\uparrow$	AMR $\downarrow$	Cov. $\uparrow$	AMR $\downarrow$	Cov. $\uparrow$	AMR $\downarrow$	Cov. $\uparrow$	AMR $\downarrow$
RDKit ETKDG	–	72.74 $\pm$ 33.18	1.04 $\pm$ 0.52	69.68 $\pm$ 37.11	1.14 $\pm$ 0.64	63.51 $\pm$ 34.74	1.15 $\pm$ 0.61	69.77 $\pm$ 38.23	1.09 $\pm$ 0.66
ASBS* (Liu et al., 2025a)	512	89.66 $\pm$ 19.42	0.86 $\pm$ 0.24	–	–	74.50 $\pm$ 32.32	1.05 $\pm$ 0.41	–	–
AS [†] (Havens et al., 2025)	256	88.60 $\pm$ 20.83	0.87 $\pm$ 0.25	59.82 $\pm$ 30.90	1.19 $\pm$ 0.37	69.37 $\pm$ 34.44	1.10 $\pm$ 0.44	40.15 $\pm$ 29.80	1.59 $\pm$ 0.54
	128	77.97 $\pm$ 29.9	0.98 $\pm$ 0.30	55.30 $\pm$ 32.88	1.24 $\pm$ 0.40	57.42 $\pm$ 35.33	1.24 $\pm$ 0.46	33.14 $\pm$ 30.71	1.75 $\pm$ 0.67
	64	29.13 $\pm$ –	1.43 $\pm$ –	20.37 $\pm$ –	1.46 $\pm$ –	5.50 $\pm$ –	1.89 $\pm$ –	4.15 $\pm$ –	1.89 $\pm$ –
Flow Sampling	256	91.89 $\pm$ 7.51	0.86 $\pm$ 0.23	62.35 $\pm$ 25.76	1.15 $\pm$ 0.35	72.61 $\pm$ 44.81	1.55 $\pm$ 0.34	44.81 $\pm$ 31.55	1.55 $\pm$ 0.46
	128	91.39 $\pm$ 16.63	0.87 $\pm$ 0.22	61.27 $\pm$ 30.55	1.18 $\pm$ 0.33	71.20 $\pm$ 34.21	1.08 $\pm$ 0.44	41.63 $\pm$ 31.80	1.61 $\pm$ 0.62
	64	90.13 $\pm$ 17.85	0.87 $\pm$ 0.23	60.75 $\pm$ 31.02	1.18 $\pm$ 0.33	71.14 $\pm$ 34.40	1.09 $\pm$ 0.43	40.82 $\pm$ 31.67	1.61 $\pm$ 0.61

Using the closed-form conditional drift derived in Section 4.1, we apply iterative Flow Sampling directly on the manifold in ambient space without reparameterizations. As shown in Figure 1, the learned diffusion accurately recovers the multimodal target density, demonstrating that Flow Sampling extends naturally beyond Euclidean spaces and can operate directly on curved geometries.

6.5. Discussion on model efficiency

Training computational cost per a single exploration, followed by a training phase Algorithm 1 is composed of three primary components: i) Number of energy gradient evaluations ii) Number of function evaluations (NFE (Train)) to sample $X_1^\theta \sim p_1^\theta$, and iii) number of optimization steps. Regarding i) i.e. number of energy gradient evaluations, AS (Havens et al., 2025), ASBS (Liu et al., 2025a) and Flow Sampling are equally efficient. Regarding iii), AS and Flow Sampling are performing 100 gradient updates per optimization stage. However, since ASBS requires fitting an auxiliary corrector network which doubles the cost of optimization and results in about $\times 2$ gradient updates per optimization stage. Most importantly, ii) the NFE to acquire a sample from the model using the Euler-Maruyama solver.

Tables 2 and 3 show that Flow Sampling maintains consistent performance across metrics even when substantially reducing NFE (Train). In particular, reducing NFE from 512 to 64 on conformer generation in Table 3 potentially cuts the exploration-phase cost by $\times 8$ relative to the 512-NFE baseline, which dominates the training cost. Furthermore, we observe strong degradation in the low-NFE regime for the baselines: ASBS on Ala2 in Table 2 and AS on conformer generation in Table 3. In contrast, Flow Sampling remains robust in both settings, indicating that the efficiency of the

exploration phase significantly improves the scalability of the method for larger models or datasets.

7. Conclusion

We introduce Flow Sampling, a principled method for learning to sample from unnormalized densities by matching conditional denoising diffusion drifts (19). We present a simplified formulation for training diffusion-based samplers, and provide a pytorch pseudo-code in Algorithm 1. We extend diffusion samplers to non-Euclidean geometries, where we provide closed-form formulas for the conditional drift on constant curvature Riemannian manifolds. We validate our Flow Sampling method empirically, observing strong results in the synthetic-energy dataset as shown in Table 1 and on small peptides in Table 2. On the amortized conformer generation benchmark reported in Table 3, our Flow Sampling demonstrates a $4\times-8\times$ speedup in simulation training, potentially enabling a considerable increase in scale of diffusion-based samplers.

Limitations Flow Sampling is trained as a fixed-point procedure: since target samples are unavailable, the ideal conditional diffusion-matching objective is approximated using samples from the detached current sampler. This is effective empirically, but we do not yet provide global convergence guarantees for the resulting replay-buffer dynamics. Finally, our Riemannian extension relies on closed-form conditional drifts for constant-curvature manifolds; extending the same efficiency to more general manifolds and constrained spaces remains an important direction.

Impact Statement

This work is primarily intended for scientific and industrial research applications and does not introduce immediate risks of misuse. Nevertheless, as with advances in generative modeling more broadly, care should be taken to ensure responsible deployment, especially in application domains such as drug discovery and materials science, where modeling biases or inaccuracies could influence downstream decisions.

Flow Sampling has direct relevance to computational chemistry, including molecular modeling and conformer generation, and may contribute to accelerating research in drug development, materials engineering, and biophysical modeling.

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A. Synthetic Energy Experiment Details

A.1. Double Well Potential (DW-4)

We use the same double-well potential as in iDEM (Akhound-Sadegh et al., 2024b), which was originally proposed in Köhler et al. (2020). DW-4 describes a pair-wise distance potential energy for a system of 4 particles $\{x_1, x_2, x_3, x_4\}$, where each particle has 2 spatial dimensions $x_i \in \mathbb{R}^2$ ($d = 8$). The potentials analytical form is given by:

$$E(x) = \frac{1}{\tau} \sum_{ij} a(d_{ij} - d_0) + b(d_{ij} - d_0)^2 + c(d_{ij} - d_0)^4, \quad d_{ij} = \|x_i - x_j\|_2 \quad (46)$$

where we set $a = 0$, $b = -4$, $c = 0.9$ and temperature $\tau = 1$.

A.2. Lennard-Jones Potential (LJ-13, LJ-55)

Similarly to DW-4, the Lennard-Jones potential is also based pair-wise distances of a system with n particles, but each particle has 3 spatial dimensions. Its analytical form is given by

$$E^{\text{LJ}}(x) = \frac{\epsilon}{\tau} \sum_{ij} \left(\left(\frac{r_m}{d_{ij}} \right)^6 - \left(\frac{r_m}{d_{ij}} \right)^{12} \right), \quad d_{ij} = \|x_i - x_j\|_2 \quad (47)$$

where r_m, τ, ϵ and c are physical constants. As in Köhler et al. (2020) and Akhound-Sadegh et al. (2024b), we add the additional a harmonic potential:

$$E^{\text{osc}}(x) = \frac{1}{2} \sum_i \|x_i - x_{\text{COM}}\|^2 \quad (48)$$

where x_{COM} refers to the center of mass of the system. Therefore, the final energy is then $E^{\text{Tot}} = E^{\text{LJ}}(x) + cE^{\text{osc}}(x)$, for c the oscillator scale. As in previous work, we use $r_m = 1$, $\tau = 1$, $\epsilon = 1$ and $c = 1.0$.

A.3. Architectures and Hyperparameters

We provide the main model and training hyperparameters used for the synthetic energy experiments in Table 4. For all three systems, we use an EGNN architecture following the setup of Akhound-Sadegh et al. (2024b). Results for iDEM are taken from the AS evaluation in (Havens et al., 2025). Each outer iteration consists of an exploration phase, where new samples are generated and their energy gradients are evaluated, followed by an optimization phase over the replay buffer.

Adaptive noise scaling. Since the magnitude of the reward-gradient signal can vary substantially across systems and during training, we adapt the diffusion coefficient γ using the current replay buffer. Let $\mathcal{B}$ denote the replay-buffer distribution over stored samples x_1 and their reward gradients. We set

$$\gamma = \frac{c}{\sqrt{\mathbb{E}_{x_1 \sim \mathcal{B}} [\|\nabla r(x_1)\|^2] + \varepsilon}}, \quad (49)$$

where c controls the scale of the reward-gradient contribution and $\varepsilon > 0$ is a small constant for numerical stability. In practice, the expectation is estimated from the stored gradients in the replay buffer. This makes γ inversely proportional to the empirical RMS reward-gradient norm, reducing sensitivity to the absolute gradient scale and to the manual clipping threshold.

A.4. Reported Metrics

Geometric $\mathcal{W}_2$ Because our energy and diffusion process is invariant to rotational and permutations symmetries, we also take these symmetries into account when measuring distance between generated and ground truth point clouds (e.g., from long run MCMC). Note this $\mathcal{W}_2$ metric is different from what is reported in Akhound-Sadegh et al. (2024b), which uses the euclidean metric.

Table 4. Hyperparameters for the synthetic energy experiments. We report the architecture, replay-buffer configuration, and optimization settings used for DW-4, LJ-13, and LJ-55.

Hyperparameter	DW-4	LJ-13	LJ-55
<i>Model</i>			
Hidden features	128	128	128
EGNN layers	3	5	5
<i>Base SDE</i>			
Harmonic source σ	1	1	1
Noise scaling γ	Adaptive, Eq. (49) ($c = 1$)	Adaptive, Eq. (49) ($c = 1$)	Adaptive, Eq. (49) ($c = 1$)
<i>Replay buffer</i>			
Buffer size	10,000	10,000	10,000
New samples per outer iteration	1,024	1,024	128
Batch size	512	512	128
<i>Training</i>			
Epochs	5,000	5,000	5,000
Iterations per Epoch	200	300	300
Learning rate	3×10^{-4}	3×10^{-4}	3×10^{-4}
Reward clipping value	100.0	100.0	100.0

$$\mathcal{W}_2(\mu, \nu) = \left(\inf_{\pi} \int \pi(x, y) d(x, y)^2 dx dy \right)^{\frac{1}{2}} \quad (50)$$

where π is the transport plan with marginals constrained to μ and ν respectively. Here the distance metric d takes into account all point-cloud symmetries, and is obtained by minimizing the squared Euclidean distance over all possible combinations of rotations, reflections ($O(d)$), and permutations ($\mathbb{S}(k)$) for k particles of spatial dimension d .

$$d(x_0, x_1) = \min_{R \in O(d), P \in \mathbb{S}(k)} \|x_0 - (R \otimes P)x_1\|_2^2. \quad (51)$$

However, computing the exact minimal squared distance is computationally infeasible in practice. Therefore, we adopt the approach of Köhler et al. (2020) and approximate the minimizer by performing a sequential search

$$d(x_0, x_1) \approx \min_{R \in SO(d)} \|x_0 - (R \otimes \tilde{P})x_1\|_2^2, \quad \tilde{P} = \arg \min_{P \in \mathbb{S}(k)} \|x_0 - Px_1\|_2^2. \quad (52)$$

Energy $\mathcal{W}_2(E(\cdot))\mathcal{W}_2$ An informative way to assess the quality of the generated samples is to look at their energy distribution with respect to the ground truth distribution obtained from energies of long-run MCMC simulations. This shows how well the generated samples are avoiding high-energy regions. Specifically, we use the Wasserstein-2 ($\mathcal{W}_2$) distance on the 1-dimensional energy distribution on $\mathbb{R}$. This results in the standard euclidean $\mathcal{W}_2$ metric

$$\mathcal{W}_2(E_\mu, E_\nu) = \left(\inf_{\pi} \int \pi(x, y) |x - y|^2 dx dy \right)^{\frac{1}{2}}, \quad (53)$$

B. Alanine Dipeptide and Tetrapeptide Experiment Details

Energy Function For both Ala2 and Ala4, we utilize a classical force-field and implicit water solvation using the OpenMM library (Eastman et al., 2017), with settings identical to (Nam et al., 2026).

Model Architecture Following the setup from (Nam et al., 2026), we employ the PaiNN architecture (Schütt et al., 2021) which is an $E(3)$ -equivariant graph neural network specifically suited for molecular energy and force-field prediction. (Nam et al., 2026) modifies the architecture to include time-conditioning, breaking parity symmetries, which restricts the symmetry to spatial $SE(3)$ -equivariance. We use the same hyper-parameters which are restated below in Table 5.

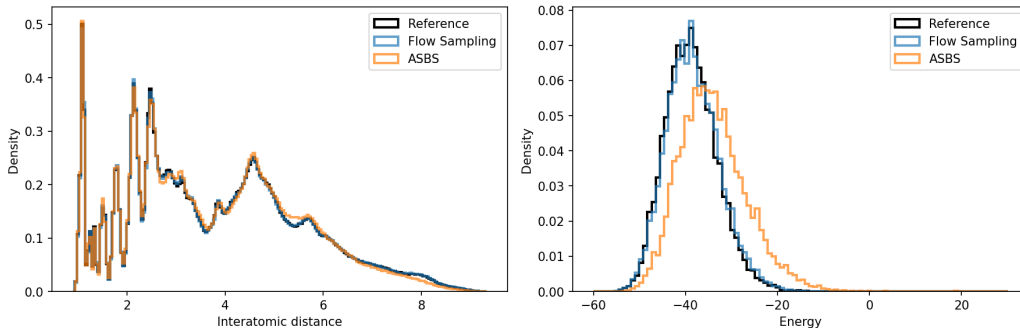


Figure 5. Ala2 interatomic distance and energy histogram over 10^4 samples evaluated against the MD data as reference.

Reference Data For Ala2, we use the same 10^7 test samples from (Midgley et al., 2023), also used by (Nam et al., 2026). These samples are generated via replica exchange MD across 21 replicas of various temperatures spanning 300K to 1300K and require 2.3×10^{10} energy evaluations to generate.

For Ala4, we directly use the test set of 5×10^7 samples generated by (Nam et al., 2026), but only subsampling 50,000 for the generated 3D scatter plot Figure 3. These samples required approximately 10^{11} energy evaluations to generate.

Ramachandran Jensen–Shannon Divergence (JSD). We evaluate the quality of generated ala2 conformations via the Jensen–Shannon divergence (JSD) of the joint backbone dihedral (ϕ, ψ) distribution. For both the generated and reference sample sets, we compute the Ramachandran angles ϕ and ψ using `mdtraj` and bin them into a 200×200 2D histogram over $[-\pi, \pi] \times [-\pi, \pi]$. Each histogram is normalized to form a discrete probability distribution:

$$\hat{p}_{ij} = \frac{h_{ij}}{\sum_{i,j} h_{ij}}, \quad (54)$$

where h_{ij} denotes the count in bin (i, j) . The JSD between the generated distribution $\hat{p}$ and the reference distribution $\hat{q}$ is then

$$\text{JSD}(\hat{p} \parallel \hat{q}) = \frac{1}{2} D_{\text{KL}}(\hat{p} \parallel m) + \frac{1}{2} D_{\text{KL}}(\hat{q} \parallel m), \quad m = \frac{1}{2}(\hat{p} + \hat{q}), \quad (55)$$

where $D_{\text{KL}}(p \parallel q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$ is the Kullback–Leibler divergence.

B.1. Interatomic Distance and Energy Histograms

In Figure 5, we provide additional qualitative comparisons on Ala2 by comparing the energy and interatomic distance distributions of generated samples against MD reference data.

C. Conformer Generation Experiment Details

C.1. Coverage Recall and Precision Metrics

Root Mean Square Deviation (RMSD) Similarly to Ganea et al. (2021) and Jing et al. (2022), we measure the so-called Average Minimum RMSD (AMR) and Coverage (COV) for Precision (P) and Recall (R). When measuring these metrics, we generate twice as many conformers as provided by CREST reference conformers. For $K = 2L$ let $\{C_l^*\}_{l \in [1, L]}$ and $\{C_k\}_{k \in [1, K]}$ be the sets of ground truth and generated conformers respectively. In our evaluations, we use $L = \max(L', 128)$, where L' is the number of reference conformers given by CREST, taking the lowest energy conformers as a subset. The RMSD metric finds the best average distance between atoms of molecule with respect to the reference molecule, taking into account all possible symmetries.

Table 5. Hyperparameters for the alanine peptide experiments. Model and training hyperparameters used for alanine dipeptide (Ala2) and alanine tetrapeptide (Ala4).

Hyperparameter	Ala2	Ala4
<i>Model</i>		
Hidden dimension	128	128
Radial basis	64	64
Control layers	4	4
Corrector layers	3	3
Radial cutoff [$\text{\AA}$]	8.0	8.0
<i>Base SDE</i>		
Harmonic Source σ	2 $\text{\AA}$	2 $\text{\AA}$
Noise scaling γ	Adaptive, Eq. (49) ($c = 1$)	Adaptive, Eq. (49) ($c = 1$)
<i>Replay buffer</i>		
Buffer size	16,384	16,384
Buffer samples per epoch	2,048	2,048
Batch size	2,048	2,048
<i>Training</i>		
Epochs	10,000	10,000
Iterations per Epoch	100	100
Initial learning rate	1×10^{-4}	1×10^{-4}
Final learning rate	3×10^{-5}	5×10^{-5}
Reward clipping value	1000.0	100.0

Coverage Recall

$$\text{COV-R}(\delta) := \frac{1}{L} |\{l \in \{1, \dots, L\} : \exists k \in \{1, \dots, K\}, \text{RMSD}(C_k, C_l^*) < \delta\}| \quad (56)$$

$$\text{AMR-R} := \frac{1}{L} \sum_{l \in \{1, \dots, L\}} \min_{k \in \{1, \dots, K\}} \text{RMSD}(C_k, C_l^*) \quad (57)$$

Coverage Precision

$$\text{COV-P}(\delta) := \frac{1}{K} |\{k \in \{1, \dots, K\} : \exists l \in \{1, \dots, L\}, \text{RMSD}(C_k, C_l^*) < \delta\}| \quad (58)$$

$$\text{AMR-P} := \frac{1}{K} \sum_{k \in \{1, \dots, K\}} \min_{l \in \{1, \dots, L\}} \text{RMSD}(C_k, C_l^*) \quad (59)$$

where $\delta > 0$ is the coverage threshold.

C.2. Hyperparameters and Architecture Details for SPICE and GEOM-DRUGS

We use the existing hyperparameter settings for AS and ASBS. Below we specify the hyperparameters specific to Flow Sampling, noting that all EGNN architectures are kept the same for all methods.

Flow Sampling: We use an Equivariant Graph Neural Network (EGNN, [Satorras et al. 2021](#)) with 12 layers and a hidden feature dimension of 128. The model is trained for 5000 outer-loop iterations, sampling 100 batches each iteration from the replay buffer. Each GPU maintains its own buffer with max size of 64000 samples and at each iteration we generate 128×8 new molecules and energy evaluations for the buffer across 8 GPUs. The model is trained with a batch size of 64 per GPU and uses a constant gamma schedule $\gamma_t \equiv 0.1$. Weight gradient clipping of 10^{20} is applied. Additionally we use the temperature $\tau = 5 \times 10^{-3}$ and a regularization constant $\alpha = 100.0$ for the regularized energy function. The regression target, the temperature-scaled gradient of the energy function is clipped at ℓ^2 norm 150.

D. Connection to Adjoint Sampling via Control Variates

In this section we relate the Adjoint Sampling ([Havens et al., 2025](#)) target to the target obtained from Flow Sampling when the supervising path is defined using the same Brownian bridge pair law. The comparison is made at the target endpoint

law. In particular, we take $X_1 \sim q$, construct the Brownian bridge interpolant used by Adjoint Sampling, and apply Flow Sampling to the conditional path induced by this interpolant. The resulting Flow Sampling target differs from the Adjoint Sampling target by a samplewise control variate whose conditional expectation given X_t is zero.

Adjoint Sampling considers the controlled SDE

$$dX_t = \sigma(t)u(X_t, t)dt + \sigma(t)dB_t, \quad X_0 = 0. \quad (60)$$

The base process is obtained by setting $u \equiv 0$, with marginals

$$p_t^{\text{base}} = \mathcal{N}(0, \nu_t I), \quad \nu_t := \int_0^t \sigma(s)^2 ds. \quad (61)$$

For a target $q(x) \propto e^{r(x)}$, the terminal cost used by Adjoint Sampling, corresponding the optimal control formulation, is

$$g(x) = \log p_1^{\text{base}}(x) - r(x). \quad (62)$$

In this zero-base-drift sampling setting, the Adjoint Sampling regression target for the control is

$$-\sigma(t)\nabla g(X_1). \quad (63)$$

Supervising Process We now define the Flow Sampling supervising path using the same Brownian bridge pair law. Given $X_1 \sim q$, let

$$X_t = \lambda_t X_1 + s_t \varepsilon, \quad \lambda_t := \frac{\nu_t}{\nu_1}, \quad s_t^2 := \frac{\nu_t(\nu_1 - \nu_t)}{\nu_1}, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (64)$$

Conditioned on ε , the map $X_1 \mapsto X_t$ is deterministic and invertible for $t \in (0, 1)$, and therefore pushes q forward to the conditional path

$$p_{t|\varepsilon}(x|\varepsilon) = \frac{1}{\lambda_t^d} q\left(\frac{x - s_t \varepsilon}{\lambda_t}\right). \quad (65)$$

Applying Flow Sampling to this conditional path gives a control target $u_{t|\varepsilon}^{\text{FS}}$. The result below shows that this target has the same conditional regression signal as the Adjoint Sampling target in (63).

Theorem D.1 (Flow Sampling recovers the Adjoint Sampling target). *Let $X_t = \lambda_t X_1 + s_t \varepsilon$ be the Brownian bridge interpolant in (64), with $X_1 \sim q$, and let $u_{t|\varepsilon}^{\text{FS}}(X_t|\varepsilon)$ be the Flow Sampling control target associated with the conditional path (65). Then, for every $t \in (0, 1)$,*

$$\mathbb{E} \left[u_{t|\varepsilon}^{\text{FS}}(X_t|\varepsilon) \mid X_t = x \right] = \mathbb{E} [-\sigma(t)\nabla g(X_1) \mid X_t = x]. \quad (66)$$

Consequently, the two samplewise targets define the same least-squares regression target as functions of X_t .

Proof. We first compute the Flow Sampling target for the conditional path (65). Along the interpolant,

$$v_{t|\varepsilon}(X_t|\varepsilon) = \dot{\lambda}_t X_1 + \dot{s}_t \varepsilon, \quad \nabla \log p_{t|\varepsilon}(X_t|\varepsilon) = \frac{1}{\lambda_t} \nabla r(X_1). \quad (67)$$

Using

$$\dot{\lambda}_t = \frac{\sigma(t)^2}{\nu_1}, \quad \dot{s}_t = \frac{\sigma(t)^2(1 - 2\lambda_t)}{2s_t}, \quad (68)$$

the Flow Sampling physical drift target is

$$b_{t|\varepsilon}^{\text{FS}}(X_t|\varepsilon) = v_{t|\varepsilon}(X_t|\varepsilon) + \frac{\sigma(t)^2}{2} \nabla \log p_{t|\varepsilon}(X_t|\varepsilon) \quad (69)$$

$$= \sigma(t)^2 \left[\frac{X_1}{\nu_1} + \frac{1 - 2\lambda_t}{2s_t} \varepsilon + \frac{1}{2\lambda_t} \nabla r(X_1) \right]. \quad (70)$$

Since the controlled SDE (60) is parameterized by the physical drift $\sigma(t)u(X_t, t)$, we compare targets at the level of the control:

$$u_{t|\varepsilon}^{\text{FS}}(X_t|\varepsilon) := \frac{1}{\sigma(t)} b_{t|\varepsilon}^{\text{FS}}(X_t|\varepsilon) \quad (71)$$

$$= \sigma(t) \left[\frac{X_1}{\nu_1} + \frac{1 - 2\lambda_t}{2s_t} \varepsilon + \frac{1}{2\lambda_t} \nabla r(X_1) \right]. \quad (72)$$

We now compare this with the Adjoint Sampling target. Since $p_1^{\text{base}} = \mathcal{N}(0, \nu_1 I)$, the terminal cost (62) satisfies

$$-\sigma(t) \nabla g(X_1) = -\sigma(t) \nabla (\log p_1^{\text{base}}(X_1) - r(X_1)) \quad (73)$$

$$= \sigma(t) \left[\frac{X_1}{\nu_1} + \nabla r(X_1) \right]. \quad (74)$$

Subtracting (74) from (72) gives

$$u_{t|\varepsilon}^{\text{FS}}(X_t|\varepsilon) - (-\sigma(t) \nabla g(X_1)) = \frac{\sigma(t)(1 - 2\lambda_t)}{2} \left[\frac{\varepsilon}{s_t} + \frac{1}{\lambda_t} \nabla r(X_1) \right]. \quad (75)$$

Thus the two targets differ pointwise by the term on the right-hand side. We now show that this term vanishes after conditioning on X_t .

Fix $X_t = x$. The conditional density of $X_1 \sim q$ given $X_t = x$ under the Brownian bridge law is given by,

$$p(X_1 = x_1 \mid X_t = x) \propto q(x_1) \exp\left(-\frac{\|x - \lambda_t x_1\|^2}{2s_t^2}\right). \quad (76)$$

Moreover, under this conditional law,

$$\varepsilon = \frac{x - \lambda_t X_1}{s_t}. \quad (77)$$

Applying Stein's identity (Hyvärinen & Dayan, 2005) to (76) gives

$$0 = \mathbb{E} [\nabla_{X_1} \log p(X_1 \mid X_t = x) \mid X_t = x] \quad (78)$$

$$= \mathbb{E} \left[\nabla r(X_1) + \frac{\lambda_t}{s_t^2} (x - \lambda_t X_1) \mid X_t = x \right] \quad (79)$$

$$= \mathbb{E} \left[\nabla r(X_1) + \frac{\lambda_t}{s_t} \varepsilon \mid X_t = x \right]. \quad (80)$$

Dividing by λ_t , we obtain

$$\mathbb{E} \left[\frac{\varepsilon}{s_t} + \frac{1}{\lambda_t} \nabla r(X_1) \mid X_t = x \right] = 0. \quad (81)$$

Combining (75) with (81) proves (66).

Finally, least-squares regression onto functions of X_t depends only on the conditional expectation of the samplewise target. Therefore (66) implies that the Flow Sampling target and the Adjoint Sampling target define the same regression target as functions of X_t . $\square$

This result is a target-law comparison. In the practical fixed-point training procedures used by both methods, the endpoint samples are obtained from the current detached sampler and are reused through a replay buffer. The calculation above explains why, when the endpoint law has reached the target distribution, the Brownian bridge Flow Sampling target reduces to the same conditional regression signal as Adjoint Sampling, with the remaining sample-wise difference acting as a zero-mean control variate.

E. Proofs

Proposition 3.1. *Let v_t be a velocity that defines the flow process (2) with a marginal p_t . Then the drift*

$$u_t(x) = v_t(x) + \frac{g_t^2}{2} \nabla \log p_t(x), \quad (12)$$

defines a diffusion process (10) with the same marginal p_t .

Proof. u_t (12) generate a diffusion process with marginal p_t , if they satisfy the Fokker–Planck equation (11). Indeed,

$$\begin{aligned} & \frac{\partial}{\partial t} p_t(x) + \nabla [p_t(x) u_t(x)] \\ &= \underbrace{\frac{\partial}{\partial t} p_t(x) + \nabla [p_t(x) v_t(x)]}_{=0, \text{ by continuity equation.}} + \frac{g_t^2}{2} \nabla [p_t(x) \nabla \log p_t(x)] \\ &= \frac{g_t^2}{2} \nabla^2 p_t(x), \end{aligned}$$

where in the first equality we substitute u_t , and in the second equality we used the assumption that v_t generate p_t , i.e., they satisfy the continuity equation (3). $\square$

Proposition 3.2. *Let $X_0 = x_0$, and $v_{t|0}$ the conditional velocity (15) that generates $p_{t|0}$ the conditional probability path (13). Then, for every $g_t \geq 0$ and $X_1 \sim q$,*

$$u_{t|0}(X_t|x_0) = \dot{\alpha}_t X_1 + \dot{\sigma}_t x_0 + \frac{g_t^2}{2\alpha_t} \nabla r(X_1). \quad (19)$$

where X_t is the interpolant (18) defined by $v_{t|0}$.

Proof of Proposition 3.2. By definition, the drift (16) $u_{t|0}$ at X_t the interpolant (18) is

$$u_{t|0}(X_t|x_0) = v_{t|0}(X_t|x_0) + \frac{g_t^2}{2} \nabla \log p_{t|0}(X_t). \quad (82)$$

We treat each term separately. The first term was also shown by Lipman et al. (2023), since the X_t the interpolant (18) is the solution to the flow defined by $v_{t|0}$ the conditional velocity (15), we have

$$v_{t|0}(X_t|x_0) = \frac{d}{dt} X_t \quad (83)$$

$$= \frac{d}{dt} (\alpha_t X_1 + \sigma_t x_0) \quad (84)$$

$$= \dot{\alpha}_t X_1 + \dot{\sigma}_t x_0, \quad (85)$$

where the first equality is by definition (2), and the second equality we substitute X_t as in equation (18).

Second, using the definition of the conditional probability path (13) $p_{t|0}$, we have

$$\nabla \log p_{t|0}(X_t|x_0) = \nabla \left(\log q \left(\frac{x - \sigma_t x_0}{\alpha_t} \right) - d \log(\alpha_t) \right) \Big|_{x=X_t} \quad (86)$$

$$= \nabla \left(r \left(\frac{x - \sigma_t x_0}{\alpha_t} \right) - \log(Z) - d \log(\alpha_t) \right) \Big|_{x=X_t} \quad (87)$$

$$= \frac{1}{\alpha_t} \nabla r \left(\frac{X_t - \sigma_t x_0}{\alpha_t} \right) \quad (88)$$

$$= \frac{1}{\alpha_t} \nabla r(X_1), \quad (89)$$

where in the second equality we substitute the definition for $q(x) = \frac{e^{r(x)}}{Z}$, and in the last equality we substitute X_t as in equation (18). $\square$

Proposition 4.1. For Riemannian manifold $\mathcal{M}$ defined in (24) with a metric Σ and constant curvature κ , the Jacobian of the geodesic w.r.t. X_1 is

$$J_t = tT_{X_1 \rightarrow X_t} P_{\dot{X}_1} + c_t(X_1, x_0) T_{X_1 \rightarrow X_t} P_{\dot{X}_1}^\perp,$$

where $T_{X_1 \rightarrow X_t} : T_{X_1} \mathcal{M} \rightarrow T_{X_t} \mathcal{M}$ is the parallel transport, $c_t(X_1, x_0)$ is a time dependent scaling,

$$c_t(X_1, x_0) = \begin{cases} \frac{\sin(t\omega_1 \sqrt{\kappa})}{\sin(\omega_1 \sqrt{\kappa})} & \text{if } \kappa > 0 \\ \frac{\sinh(t\omega_1 \sqrt{|\kappa|})}{\sinh(\omega_1 \sqrt{|\kappa|})} & \text{if } \kappa < 0 \end{cases}, \quad (35)$$

and $\omega_1 = \|\dot{X}_1\|_\Sigma$ the geodesic distance of X_1 and x_0 .

Proof of Proposition 4.1. Our proof closely follows results about Jacobi fields from the book Introduction to Riemannian Manifolds (Lee, 2018).

Denote the geodesic starting at X_1 and ending at x_0 by

$$\gamma(s) \in \mathcal{M}, \forall s \in [0, 1] \quad \gamma(0) = X_1, \quad \gamma(1) = x_0. \quad (90)$$

The velocity of the geodesic is defined as $\gamma'(s) = \frac{d}{ds}\gamma(s)$, and for every $s \in [0, 1]$, $\gamma'(s) \in T_{\gamma(s)} \mathcal{M}$. Indeed

$$\langle \gamma'(s), \gamma(s) \rangle_\Sigma = \left\langle \frac{d}{ds}\gamma(s), \gamma(s) \right\rangle_\Sigma \quad (91)$$

$$= \frac{1}{2} \left(\left\langle \frac{d}{ds}\gamma(s), \gamma(s) \right\rangle_\Sigma + \left\langle \gamma(s), \frac{d}{ds}\gamma(s) \right\rangle_\Sigma \right) \quad (92)$$

$$= \frac{1}{2} \frac{d}{ds} \langle \gamma(s), \gamma(s) \rangle_\Sigma = 0. \quad (93)$$

The covariant derivative D_s on our embedded manifold can be understood as $D_s v = P_{\gamma(s)}^\perp \frac{d}{ds} v(s)$ for any field $v(s) \in T_{\gamma(s)} \mathcal{M}$ along the geodesic γ . A field $v(s) \in T_{\gamma(s)} \mathcal{M}$ is said to be parallel along $\gamma(s)$ if $D_s v \equiv 0$. By definition of the geodesic, the velocity $\gamma'(s)$ is parallel along $\gamma(s)$, i.e.,

$$D_s \gamma' \equiv 0 \quad (94)$$

Every parallel field $v(s)$ along $\gamma(s)$ given by an isometric linear map $T_{\gamma(0) \rightarrow \gamma(s)} : T_{\gamma(0)} \mathcal{M} \rightarrow T_{\gamma(s)} \mathcal{M}$ called the *parallel transport* along γ and an initial condition,

$$v(s) = T_{\gamma(0) \rightarrow \gamma(s)} v(0). \quad (95)$$

For our set of manifolds (24), for any $X_1 \in \mathcal{M}$ and $x_0 \in \mathcal{M}$ with a unique minimizing geodesic,

$$\gamma(s) = \exp_{X_1}(sV_1), \quad V_1 = \log_{X_1}(x_0), \quad (96)$$

where $\exp_{X_1} : T_{X_1} \rightarrow \mathcal{M}$ and $\log_{X_1} : \mathcal{M} \rightarrow T_{X_1}$ are the exponential and logarithmic map of the manifold. By definition of the exponential map

$$\gamma'(0) = V_1 \quad (97)$$

Note, both the basis of $\exp_{X_1}$ and $\log_{X_1}$ depends on X_1 , hence the Jacobian of geodesic requires the total variation w.r.t. X_1 .

$$D_{X_1} \gamma(s) = \underbrace{(d_x \exp_{X_1})_{sV_1}}_{\text{Base-point variation}} + \underbrace{(d_v \exp_{X_1})_{sV_1} \circ (sD_{X_1} \log_{X_1}(x_0))}_{\text{Argument variation}} \quad (98)$$

To compute the variations of the exponent map, we use tools of Jacobi fields. A field $j(s)$ along $\gamma(s)$ is said to be a Jacobi field if it satisfies the Jacobi equation. In constant curvature manifolds, the Jacobi equation takes the simple form

$$D_s^2 j(s) + \kappa [\langle \gamma'(s), \gamma'(s) \rangle_\Sigma j(s) - \langle \gamma'(s), j(s) \rangle_\Sigma \gamma'(s)] = 0, \quad (99)$$

Proposition 10.1 (Lee, 2018) states that every variation through the geodesic is a Jacobi field. That is, for every vector $v_1 \in T_{X_1}\mathcal{M}$, both

$$j_x(s) = (d_x \exp_{X_1})_{sV_1} \cdot v_1, \quad \text{and} \quad j_v(s) = (d_v \exp_{X_1})_{sV_1} \cdot v_1 \quad (100)$$

are Jacobi fields, where “ $\cdot$ ” denotes the action of the variation on v_1 .

Additionally, Proposition 10.2 (Lee, 2018) states that for every $s_0 \in [0, 1]$ and pair of vectors $v, w \in T_{\gamma(s_0)}\mathcal{M}$ there exists a unique Jacobi field $j(s)$ along gamma such that

$$j(s_0) = v \quad D_s j(s_0) = w. \quad (101)$$

We use the uniqueness to compute the variations $(d_x \exp_{X_1})_{sV_1}$ and $(d_v \exp_{X_1})_{sV_1}$. That is, we have that at $s = 0$, for every $v_1 \in T_{X_1}\mathcal{M}$,

$$(d_x \exp_{X_1})_0 \cdot v_1 = v_1, \quad D_s(d_x \exp_{X_1})_0 \cdot v_1 = 0, \quad \text{and} \quad (d_v \exp_{X_1})_0 \cdot v_1 = 0, \quad D_s(d_v \exp_{X_1})_0 \cdot v_1 = v_1. \quad (102)$$

Hence by uniqueness it is suffice to find two Jacobi fields $j_x(s)$ and $j_v(s)$ along $\gamma(s)$ that satisfy these boundary conditions.

A field $v^\parallel(s) \in T_{\gamma(s)}\mathcal{M}$ is called *tangential field* if $v^\parallel(s) \in \text{span}\{\gamma'(s)\}$ for every $s \in [0, 1]$, and a field $v^\perp(s) \in T_{\gamma(s)}\mathcal{M}$ is called *normal field* if $\langle v(s), \gamma'(s) \rangle_\Sigma = 0$ for every $s \in [0, 1]$. We show that the Jacobian can be decomposed to a tangential field and a normal field. Importantly, the Jacobi equation (99) takes a simpler form in case $j^\parallel(s)$ is a tangential field or $j^\perp$ is a normal field. Indeed

$$D_s^2 j^\parallel(s) = 0, \quad \text{and} \quad D_s^2 j^\perp(s) + \kappa \langle \gamma'(s), \gamma'(s) \rangle_\Sigma j^\perp(s) = 0. \quad (103)$$

Additionally, since $\gamma'(s)$ is parallel along γ ,

$$\langle \gamma'(s), \gamma'(s) \rangle_\Sigma = \langle T_{\gamma(0) \rightarrow \gamma(s)} \gamma'(0), T_{\gamma(0) \rightarrow \gamma(s)} \gamma'(0) \rangle_\Sigma \quad (104)$$

$$= \langle \gamma'(0), \gamma'(0) \rangle_\Sigma \quad (105)$$

$$= \langle V_1, V_1 \rangle_\Sigma \quad (106)$$

$$= \omega_1^2, \quad (107)$$

where in the second equality used that the parallel transport $T_{\gamma(0) \rightarrow \gamma(s)}$ is an isometry. Thus the Jacobi equation (99) further simplify to

$$D_s^2 j^\parallel(s) = 0, \quad \text{and} \quad D_s^2 j^\perp(s) + \kappa \omega_1^2 j^\perp(s) = 0. \quad (108)$$

The simplest way to generate fields that conserve orientation along $\gamma(s)$, i.e., tangential or normal are using the parallel transport. Since $T_{X_1 \rightarrow \gamma(s)}$ is an isometry if $v_1^\parallel, v_1^\perp \in T_{X_1}\mathcal{M}$ are tangential and normal vectors (reps.), i.e., $v_1^\parallel \in \text{span}\{V_1\}$ and $\langle v_1^\perp, V_1 \rangle_\Sigma = 0$, then $v_1^\parallel$ and $v_1^\perp$ give rise to a tangential and normal (resp.) parallel field along $\gamma(s)$ defined as

$$v^\parallel(s) = T_{X_1 \rightarrow \gamma(s)} v_1^\parallel, \quad \text{and} \quad v^\perp(s) = T_{X_1 \rightarrow \gamma(s)} v_1^\perp. \quad (109)$$

Indeed,

$$v^\parallel(s) = T_{X_1 \rightarrow \gamma(s)} v_1^\parallel \in \text{span}\{T_{X_1 \rightarrow \gamma(s)} V_1\} = \text{span}\{\gamma'(s)\}, \quad (110)$$

and

$$\langle v^\perp(s), \gamma'(s) \rangle_\Sigma = \langle T_{X_1 \rightarrow \gamma(s)} v_1^\perp, T_{X_1 \rightarrow \gamma(s)} V_1 \rangle_\Sigma = \langle v_1^\perp, V_1 \rangle_\Sigma = 0. \quad (111)$$

First we consider the tangential direction. Let $v_1^\parallel \in \text{span}\{V_1\}$, and consider $j_x^\parallel(s) = v^\parallel(s)$ and $j_v^\parallel(s) = s v^\parallel(s)$, both tangential fields and since $v^\parallel(s)$ is parallel $D_s v^\parallel \equiv 0$. Thus both $j_x^\parallel(s)$ and $j_v^\parallel(s)$ satisfy the Jacobi equation for the tangential case 108, and the initial conditions at $s = 0$ by construction are

$$j_x^\parallel(0) = v_1^\parallel, \quad D_s j_x^\parallel(0) = 0, \quad \text{and} \quad j_v^\parallel(0) = 0, \quad D_s j_v^\parallel(0) = v_1^\parallel. \quad (112)$$

Hence uniqueness implies,

$$j_x^\parallel(s) = (d_x \exp_{X_1})_{sV_1} \cdot v_1^\parallel, \quad \text{and} \quad j_v^\parallel(s) = (d_v \exp_{X_1})_{sV_1} \cdot v_1^\parallel. \quad (113)$$

Second we consider the normal directions. Let $v_1^\perp \in T_{X_1}\mathcal{M}$ be such that $\langle v_1^\perp, V_1 \rangle_\Sigma = 0$. and consider $j^\perp(s) = c_s(X_1, x_0)v^\perp(s)$, where $c(X_1, x_0) : [0, 1] \rightarrow \mathbb{R}$ is some smooth scaling function. Then since $v^\perp(s)$ is parallel field along $\gamma(s)$, i.e., $D_s v^\perp \equiv 0$, the normal case of the Jacobi equation (108) reduce to

$$c_s''(X_1, x_0)v^\perp(s) + \kappa w_1^2 c_s(X_1, x_0)v^\perp(s) = 0, \quad (114)$$

which is true if and only if $c_s(X_1, x_0)$ satisfies the ODE,

$$c_s''(X_1, x_0) + \kappa w_1^2 c_s(X_1, x_0) = 0. \quad (115)$$

The solutions to the ODE in equation (115) are

$$c_s(X_1, x_0) = \begin{cases} c_0(X_1, x_0) \cos(\sqrt{\kappa}\omega_1 s) + \frac{c_0'(X_1, x_0)}{\sqrt{\kappa}\omega_1} \sin(\sqrt{\kappa}\omega_1 s) & \text{if } \kappa > 0 \\ c_0(X_1, x_0) \cosh(\sqrt{|\kappa|}\omega_1 s) + \frac{c_0'(X_1, x_0)}{\sqrt{|\kappa|}\omega_1} \sinh(\sqrt{|\kappa|}\omega_1 s) & \text{if } \kappa < 0 \end{cases}, \quad (116)$$

where $c_0(X_1, x_0)$ and $c_0'(X_1, x_0)$ are determined by the boundary conditions at $s = 0$. Thus we define,

$$j_x^\perp(s) = c_s^x(X_1, x_0)v^\perp(s), \quad \text{and} \quad j_v^\perp(s) = c_s^v(X_1, x_0)v^\perp, \quad (117)$$

where

$$c_s^x(X_1, x_0) = \begin{cases} \cos(\sqrt{\kappa}\omega_1 s) & \text{if } \kappa > 0 \\ \cosh(\sqrt{|\kappa|}\omega_1 s) & \text{if } \kappa < 0 \end{cases}, \quad (118)$$

and

$$c_s^v(X_1, x_0) = \begin{cases} \frac{1}{\sqrt{\kappa}\omega_1} \sin(\sqrt{\kappa}\omega_1 s) & \text{if } \kappa > 0 \\ \frac{1}{\sqrt{|\kappa|}\omega_1} \sinh(\sqrt{|\kappa|}\omega_1 s) & \text{if } \kappa < 0. \end{cases} \quad (119)$$

Then they satisfy the initial conditions

$$j_x^\perp(0) = v_1^\perp, \quad D_s j_x^\perp(0) = 0, \quad \text{and} \quad j_v^\perp(0) = 0, \quad D_s j_v^\perp(0) = v_1^\perp. \quad (120)$$

Hence uniqueness implies,

$$j_x^\perp(s) = (d_x \exp_{X_1})_{sV_1} \cdot v_1^\perp, \quad \text{and} \quad j_v^\perp(s) = (d_v \exp_{X_1})_{sV_1} \cdot v_1^\perp. \quad (121)$$

Let $v_1 \in T_{X_1}\mathcal{M}$, finally we can write the full Jacobi fields for the variations of the exponent, i.e.,

$$j_x(s) = (d_x \exp_{X_1})_{sV_1} \cdot v_1, \quad \text{and} \quad j_v(s) = (d_v \exp_{X_1})_{sV_1} \cdot v_1. \quad (122)$$

Define $v_1^\parallel = P_{V_1} v_1$ and $v_1^\perp = P_{V_1}^\perp v_1$, where P_{V_1} is the orthogonal projection (27) and $P_{V_1}^\perp$ its orthogonal compliment. Then $v_1^\parallel$ is tangential vector and $v_1^\perp$ is normal vector, thus they give rise to Jacobi fields $j_x^\parallel, j_v^\parallel$ which satisfy the tangential Jacobi equation (108), and $j_x^\perp, j_v^\perp$ which satisfy the normal Jacobi equation (108). Lastly, since the Jacobi equation (99) decomposes to two independent linear differential equations (i.e., one tangential and one normal) as in equation (108), the fields

$$j_x(s) = j_x^\parallel(s) + j_x^\perp(s) \quad (123)$$

$$= T_{X_1 \rightarrow \gamma(s)} P_{V_1} v_1 + c_s^x(X_1, x_0) T_{X_1 \rightarrow \gamma(s)} P_{V_1}^\perp v_1 \quad (124)$$

$$= [T_{X_1 \rightarrow \gamma(s)} P_{V_1} + c_s^x(X_1, x_0) T_{X_1 \rightarrow \gamma(s)} P_{V_1}^\perp] v_1 \quad (125)$$

and

$$j_v(s) = j_v^\parallel(s) + j_v^\perp(s) \quad (126)$$

$$= s T_{X_1 \rightarrow \gamma(s)} P_{V_1} v_1 + c_s^v(X_1, x_0) T_{X_1 \rightarrow \gamma(s)} P_{V_1}^\perp v_1 \quad (127)$$

$$= [s T_{X_1 \rightarrow \gamma(s)} P_{V_1} + c_s^v(X_1, x_0) T_{X_1 \rightarrow \gamma(s)} P_{V_1}^\perp] v_1, \quad (128)$$

are Jacobi fields and they satisfy boundary condition (102).

As last step, we compute the variation of the logarithmic $D_{X_1} \log_{X_1}(x_0)$. By definition

$$x_0 = \gamma(1) = \exp_{X_1}(\log_{X_1}(x_0)). \quad (129)$$

Since x_0 is fixed,

$$0 = D_{X_1} \gamma(1) = (d_x \exp_{X_1})_{V_1} + (d_v \exp_{X_1})_{V_1} \circ D_{X_1} \log_{X_1}(x_0) \quad (130)$$

$$= [T_{X_1 \rightarrow x_0} P_{V_1} + c_1^x(X_1, x_0) T_{X_1 \rightarrow x_0} P_{V_1}^\perp] + [T_{X_1 \rightarrow x_0} P_{V_1} + c_1^v(X_1, x_0) T_{X_1 \rightarrow x_0} P_{V_1}^\perp] \circ D_{X_1} \log_{X_1}(x_0) \quad (131)$$

$$= T_{X_1 \rightarrow x_0} \circ [(P_{V_1} + c_1^x(X_1, x_0) P_{V_1}^\perp) + (P_{V_1} + c_1^v(X_1, x_0) P_{V_1}^\perp) \circ D_{X_1} \log_{X_1}(x_0)] \quad (132)$$

$$= T_{X_1 \rightarrow x_0} \circ [(P_{V_1} + P_{V_1} \circ D_{X_1} \log_{X_1}(x_0)) + (c_1^v(X_1, x_0) P_{V_1}^\perp + c_1^v(X_1, x_0) P_{V_1}^\perp \circ D_{X_1} \log_{X_1}(x_0))] . \quad (133)$$

Since the parallel transport is invertible, this implies

$$P_{V_1} \circ D_{X_1} \log_{X_1}(x_0) = -P_{V_1}, \quad \text{and} \quad P_{V_1}^\perp \circ D_{X_1} \log_{X_1}(x_0) = -\frac{c_1^x(X_1, x_0)}{c_1^v(X_1, x_0)} P_{V_1}^\perp \quad (134)$$

We conclude that the Jacobian $D_{X_1} \gamma(s)$ is

$$D_{X_1} \gamma(s) = (d_x \exp_{X_1})_{sV_1} + (d_v \exp_{X_1})_{sV_1} \circ D_{X_1} \log_{X_1}(x_0) \quad (135)$$

$$= [T_{X_1 \rightarrow x_0} P_{V_1} + c_s^x(X_1, x_0) T_{X_1 \rightarrow \gamma(s)} P_{V_1}^\perp] + [sT_{X_1 \rightarrow \gamma(s)} P_{V_1} + c_s^v(X_1, x_0) T_{X_1 \rightarrow \gamma(s)} P_{V_1}^\perp] \circ D_{X_1} \log_{X_1}(x_0) \quad (136)$$

$$= T_{X_1 \rightarrow \gamma(s)} [P_{V_1} + sP_{V_1} \circ D_{X_1} \log_{X_1}(x_0)] + T_{X_1 \rightarrow \gamma(s)} [c_s^x(X_1, x_0) P_{V_1}^\perp + c_s^v(X_1, x_0) P_{V_1}^\perp \circ D_{X_1} \log_{X_1}(x_0)] \quad (137)$$

$$= (1-s) T_{X_1 \rightarrow \gamma(s)} P_{V_1} + \frac{c_1^v(X_1, x_0) c_s^x(X_1, x_0) - c_1^x(X_1, x_0) c_s^v(X_1, x_0)}{c_1^v(X_1, x_0)} T_{X_1 \rightarrow \gamma(s)} P_{V_1}^\perp \quad (138)$$

$$= (1-s) T_{X_1 \rightarrow \gamma(s)} P_{V_1} + c_{1-s}(X_1, x_0) T_{X_1 \rightarrow \gamma(s)} P_{V_1}^\perp, \quad (139)$$

where

$$c_{1-s}(X_1, x_0) = \begin{cases} \frac{\sin((1-s)\omega_1\sqrt{\kappa})}{\sin(\omega_1\sqrt{\kappa})} & \text{if } \kappa > 0 \\ \frac{\sinh((1-s)\omega_1\sqrt{|\kappa|})}{\sinh(\omega_1\sqrt{|\kappa|})} & \text{if } \kappa < 0 \end{cases}, \quad (140)$$

Finally, noting that the geodesic interpolant (30) is $X_t = \gamma(1-t)$, which implies $V_1 = -\dot{X}_1$, and therefore $\text{span}\{V_1\} = \text{span}\{\dot{X}_1\}$, completes the proof. $\square$